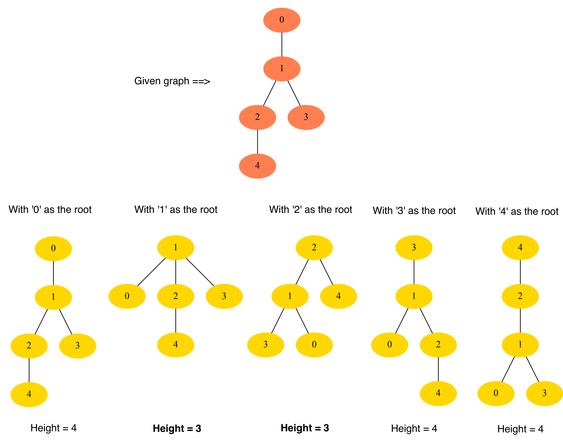
Problem Challenge 2: Minimum Height Trees (hard)

Problem Statement

We are given an undirected graph that has the characteristics of a k-ary tree. In such a graph, we can choose any node as the root to make a k-ary tree. The root (or the tree) with the minimum height will be called Minimum Height Tree (MHT). There can be multiple MHTs for a graph. In this problem, we need to find all those roots which give us MHTs. Write a method to find all MHTs of the given graph and return a list of their roots.

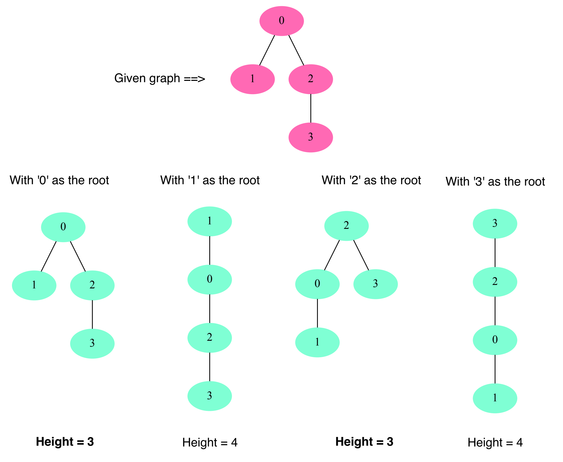
Example 1:

```
Input: vertices: 5, Edges: [[0, 1], [1, 2], [1, 3], [2, 4]]
Output:[1, 2]
Explanation: Choosing '1' or '2' as roots give us MHTs. In the below
diagram, we can see that the height of the trees with roots '1' or '2'
is three which is the minimum.
```



Example 2:

```
Input: vertices: 4, Edges: [[0, 1], [0, 2], [2, 3]]
Output:[0, 2]
Explanation: Choosing '0' or '2' as roots give us MHTs. In the below
diagram, we can see that the height of the trees with roots '0' or '2'
is three which is minimum.
```



Example 3:

```
Input: vertices: 4, Edges: [[0, 1], [1, 2], [1, 3]]
Output:[1]
```

Try it yourself

Try solving this question here:

```java
import java.util.*;

class Solution {
  public List<Integer> findTrees(int nodes, int[][] edges) {
    List<Integer> minHeightTrees = new ArrayList<>();
    // TODO: Write your code here
    return minHeightTrees;
  }
}
```

Test Case | Run Code Result | Stdout

```
5
[[0,1],[1,2],[1,3],[2,4]]
```

Use Example Testcases | Run | Submit